A Scrupulous Analysis of the Ethical & Moral Implications of Self-Driving Cars

Lindsay R. Rogers

School of Information, University of South Florida

LIS 4414 – Information Policy & Ethics

Dr. John Gathegi

June 16, 2025

A Scrupulous Analysis of the Ethical & Moral Implications of Self-Driving Cars

As the modern world moves increasingly toward artificial intelligence and automation, automotive manufacturers, governments, and consumers are looking ahead to the future of driving. Fully self-driving cars have emerged as an ambitious goal for the future. The aim of this technology is to make driving and transportation safer by eliminating risks related to human errors. However, implementations of self-driving car technology in the last decade have shown that autonomous vehicles come with their own risks as well as new considerations. In a TED talk given by Patrick Lin, the speaker presents a thought experiment scenario in which an autonomous car must make a practical decision with ethical implications. In the scenario, a car is traveling on a busy highway when a "large, heavy object falls off the truck in front of [it]" (Lin, 2015). The car cannot stop, so its only other options are to choose to hit the object and endanger the driver, hit a nearby SUV on the left, damaging both cars, or hit a nearby motorcycle on the right, prioritizing the driver's life at the expense of the motorcyclist's. In the context of computerized programming instructions which are set in advance by humans, this instance becomes a premeditated moral choice rather than an instinctual reaction (Lin, 2015). This raises important ethical questions about how autonomous cars should be programmed to behave when handling accident scenarios, as well as who is responsible for regulating those behaviors. In this essay, both the practical and ethical implications of self-driving cars will be explored.

Perspectives and Case Studies on Self-Driving Cars

With the push for growth and development of self-driving technology, the National Highway Traffic Safety Administration (NHTSA) has put out guidance on standard levels of automation and what they entail. In the web article *Automated Vehicles for Safety*, the NHTSA delves into the history and development of autonomous cars, classifying six distinct levels of

automation. Level zero offers some convenience features like breaking assistance or collision warnings but leaves the human driver fully in control. Level one is described as "driver assistance," wherein the car's system assists the driver seamlessly through methods like adaptive cruise control or lane retention. Level two expands this concept further with continuous driver assistance, leaving the driver in control of most features while offering some additional autonomous features like steering assistance or highway autopilot. At level three, conditional automation is introduced, meaning that "[the] [s]ystem actively performs driving tasks while [the] driver remains available to take over" (NHTSA, 2021). The NHTSA notes that autonomous technology of this level, as well as succeeding levels, are not yet very common. At level four, high automation is reached, where "the system handles all driving tasks" but "in limited service areas." Finally, at level five, the autonomous system is fully in control in all circumstances and any occupants are merely passengers. (NHTSA, 2021). Given that most cars which come with autonomous features are at level three or less, more time, data, and advances in self-driving technology are needed.

But adopting this kind of new technology in a layman's world is no easy feat, especially for non-technical policymakers. The Association for Computing Machinery (ACM) is an organization that plays an important role in the public policymaking process relevant to self-driving cars. ACM's US Technology Policy Committee (USTPC) works directly with institutions in the U.S. government by "regularly [educating] and [informing] Congress, the Administration, and the courts about significant developments in the computing field and how those developments affect public policy in the United States" ("ACM Provides policymakers", n.d.). The organization identifies some of the most technologically relevant issues today as digital privacy, AI, cybersecurity, and ethics, among others. Their efforts help to educate non-technical

legislators on modern technology related issues so that they can make better informed policy decisions ("ACM Provides policymakers", n.d.). Having this kind of resource to provide expert advice to policymakers is essential for the rapidly developing technological landscape of today.

As lawmakers embrace the push for self-driving cars, some U.S. states have already begun to accommodate their introduction. Author Mark Harris, writing for IEEE, goes into depth on the state of California's decision to "green light" self-driving cars. With the interest in hastening electric vehicle technology development, California opted in 2017 to make two major policy changes: to allow self-driving cars to operate without a licensed driver inside, and to allow "[electric vehicle] manufacturers to certify their own fully driverless vehicles" (Harris, 2017). According to Harris, this decision omitted time consumption from previously required inspections. The manufacturers would also no longer need to request permission from cities where their self-driving cars are operated. Instead, the manufacturers need only notify the city that they will be present (2017). To mitigate the risks associated with powerless humans riding in self-driving cars, autonomous vehicle manufacturers will still be required to meet certain safety standards. For instance, "the DMV (…) requires all such autonomous vehicles to have a dedicated communications link with a remote operator (…) [who] could even teleoperate the car in an emergency" (Harris, 2017). This example shows California's attempt to aid the progress of self-driving car development while balancing the need to maintain human safety.

It follows that a question is inherent to self-driving car technology: what regulations and policy changes are necessary to protect manufacturers and consumers? Author John Villasenor, writing for Brookings, presents an argument for the case that products liability laws are already capable of sufficiently handling concerns regarding self-driving cars. The author argues that attempting to introduce new legislation may be unneeded and only serve to slow down

autonomous car development, stating “statutes aimed at protecting the manufacturers of autonomous vehicle technology are unnecessary” (Villasenor, 2014). Villasenor posits that liability laws which already exist have been applied to similar cases such as automated planes and traditional motor vehicles. He states that “there is no reason to expect that the legal system will be unable to resolve [cases involving] (…) future vehicle automation systems” (Villasenor, 2014).

However, it may be the case that some policy updates have been warranted. In a Brookings article written by author Jack Karsten, the relationship between safe, autonomous car policy and public adoption is explored. In the article, the author evaluates a paper by Darrell West which considers both the regulatory challenges and possible benefits of self-driving cars following the release of “long-awaited policy guidance for autonomous vehicles” in 2016 (Karsten, 2016). These policy changes are critical for the future of self-driving cars. The updates, put forth by the U.S. Department of Transportation and NHTSA, attempted to “harmonize rules for states and driverless car developers in a way that promotes both public safety and innovation” (Karsten, 2016). Karsten notes that the proliferation of self-driving cars is expected to create significant economic value, improve road safety, and reduce traffic congestion. To achieve this, improvements in both topical and digital infrastructure as well as “public acceptance” will be crucial (Karsten, 2016). Indeed, safety may be a primary key to fostering public trust, which is an important precursor to the adoption of self-driving cars.

It seems pertinent, then, that self-driving cars be supervised carefully as they work to pursue the ultimate goal of safe, full autonomy. In a Scientific American article, author Jeremy Hsu presents the argument that successful automotive autonomy does not necessarily equate to autonomous maturity and safety. Citing information scientist Nidhi Kalra, Hsu compares self-

driving cars to the intellectual developmental stage of human teenagers, suggesting that self-driving cars still lack practical experience and are in need of supervision while they learn. Hsu points out that while there is a need for standardized safety with self-driving cars, "safety" itself is difficult to unanimously define. He reasons that additional research will be an important part of future development for self-driving cars if they are ever to reach "full autonomy" (Hsu, 2017). Indeed, current autonomous driving technology is still quite limited. It seems evident that more time, research, data collection, and careful supervision are needed to perfect the technology.

The judgmental effectiveness of self-driving cars will be another highly important part of their safe implementation. Author Peter Fairley, writing for IEEE Spectrum, addresses the fundamental issues with self-driving cars operating concurrently with bicycles on public roads. Most prevalent among these are that bikes are small and prone to sudden, unpredictable movements, are inconsistent in shape and size, and most importantly, the algorithms on which these vehicles operate are not sufficiently trained on data involving bicycles. Fairley notes in the article that "[these] systems train themselves by studying thousands of images in which known objects are labeled (…) [but] most of the training has concentrated on images featuring cars, with far fewer bikes" (2017). It is clear to a human spectator that bicycle behaviors are very different than the typical behaviors of cars, but this distinction remains unclear to existing self-driving algorithms. As evidence of this safety gap, Fairley gives the example of the Deep3DBox self-driving algorithm, which "[identified] 89 percent of cars (…) [but] only 74 percent of bikes in [a] benchmarking test (…) conducted by researchers at George Mason University" (2017). The algorithm's object orientation scores were even less promising at "88 percent [for] cars" and "59 percent [of] bikes" (Fairley, 2017). It follows that self-driving cars cannot be safely implemented

without further training on important data beyond responding to other cars, as in the case with bicycles.

Writing again for Brookings, author Jack Karsten details another blind spot when it comes to semi-autonomous vehicles. He examines a report by the NHTSA about a fatal crash in a semi-autonomous Tesla vehicle while it was operating on autopilot. According to the report, the system was operating normally at the time of the fatality. Karsten cites "driver inattention" as a prevailing problem for semi-autonomous cars, noting that "[s]emi autonomous features (…) improve vehicle safety so long as drivers continue to pay attention when vehicles or other objects suddenly enter their path" (2017). It is for this reason that the author suggests further improvements in monitoring both road conditions and the conditions of the human driver inside the car are necessary to improve safety measures (Karsten, 2017). Semi-autonomous driving systems, which still primarily rely on human intelligence with a minor degree of computer assistance, would do well to implement more advanced monitoring of the human driver.

An Ethical Analysis of Self-Driving Cars

It seems evident that when handing control over to self-driving cars, there are many ethical implications not only for safe operation, but particularly for instances of accident handling. Many traditional car accidents do occur because of human error or lapses in human judgment, but others result from the uncontrolled chaos of nature. Car manufacturers have incentives to make autonomous driving technology as safe as possible, but even the most precisely programmed machine cannot anticipate every random event (Villasenor, 2014). Until recently, any actions one takes in an emergency situation like a car accident are not pre-meditated. But self-driving cars must be programmed to make concrete choices in emergency situations, and these choices become inherently moral when humans write code to take control

(Lin, 2015). Automotive manufacturers and governments must come together to decide what ethical choices should be in the event of accidents involving self-driving cars. An immediate implication of this is not only to decide to prioritize life over property, but also to select for whose life is worth prioritizing. Should an autonomous car decide to save its occupants or occupants with a better chance of survival in another vehicle (Lin, 2015)? What happens if it makes the wrong choices? Who is then responsible or liable?

If self-driving car manufacturers can each make their own ethical choices in this regard, then an even more complex iteration of algorithmic bias could be introduced. In a TED talk given by Eli Pariser related to personalized social media, the speaker cautions for awareness of algorithmic echo chambers. Just as social media and web search sites are algorithmically filtering what different users see, self-driving algorithms could also produce different results for different users (Pariser, 2025). But in the context of self-driving cars, different results could mean life or death. It is common knowledge in the study of ethics that different ethical frameworks, which use different manners of reasoning, often conflict. For instance, what an act utilitarian and what a Kantian deem as a “right” choice may be completely opposite in certain circumstances. In the event that different autonomous car manufacturers use different ethical approaches to accident handling, then it is very likely that they would also produce different outcomes based on their programming. Human programmers themselves could potentially contribute to bias in their coded decisions, whether consciously or not. These decisions could then influence who lives or who dies in a self-driving car accident. Just as the unseen personalized algorithms of social media can distort our worldview, the algorithms that govern the moral choices of self-driving cars could carry biases that remain hidden from consumers (Pariser, 2025). An example of this already exists as in the case presented by Fairley, which unintentionally prioritized identifying

cars over bicycles due to limited data (2017). These judgments do not even have to be intentional to incur a moral burden when lives are at risk.

In the future, these decisions will have to be standardized by industries and governments while being implemented by programmers and manufacturers. If some sort of moral standard is not provided, then consumers riding in different brands of cars may have very different survival outcomes in accident scenarios. This is ripe with potential for lawsuits that neither businesses nor consumers would want to face unnecessarily. The burden would then lay on computer programmers and businesses to uphold these standards. In many cases, this would literally involve writing code to determine who would likely live or die during an accident. It would be impractical and immoral to place this burden of moral judgment on the backs of programmers who would control the fate of all of a commuting society. Instead, the general population should have the opportunity to vote ethical standards into law for self-driving cars which can then be upheld by governments. One moral framework from which to begin approaching this necessary discussion could be by implementing principles of act utilitarianism. This moral theory prioritizes benefits to the many and attempts to reduce suffering. An interpretation of act utilitarianism in the context of a self-driving car accident might be to take the course of action that saves the most lives. However, this should not be a broad and sweeping rule, as the complexity of accident scenarios requires more nuanced decision making. This moral decision making will be a significant challenge for the future.

Conclusion

As self-driving cars continue to become an inevitable part of the future, they bring with them both practical and ethical implications beyond the scope of traditional cars. Implementing safety measures that protect both consumers and manufacturers is as critical to the proliferation

of self-driving cars as public trust is. Both professional institutions and policy already exist which can help to shape autonomous vehicle governance in the future. For the time being, self-driving technology remains in its early stages. More data, more testing, and further considerations will be required in order to bring this technology to its full potential. Even given all of these improvements, there are still inherent moral implications with this technology. When humans take control of situations otherwise out of their control, such as in the case with car accidents, the issue becomes an intentional decision rather than a chance of nature. People will need to come together with governments and self-driving car manufacturers in order to decide what kind of moral choices self-driving cars should make, because those choices will one day determine who lives and who dies.

References

ACM provides policymakers with independent, nonpartisan, and technology-neutral resources. (n.d.). Www.acm.org. https://www.acm.org/public-policy

Fairley, P. (2017, February 24). *Self-driving cars have a bicycle problem*. IEEE Spectrum. https://spectrum.ieee.org/selfdriving-cars-have-a-bicycle-problem

Harris, M. (2017, March 10). *California gives the green light to self-driving cars*. IEEE Spectrum. https://spectrum.ieee.org/california-gives-the-green-light-to-selfdriving-cars

Hsu, J. (2017, January 18). *When it comes to safety, autonomous cars are still "teen drivers."* Scientific American. https://www.scientificamerican.com/article/when-it-comes-to-safety-autonomous-cars-are-still-teen-drivers1/

Karsten, J. (2016, September 20). *New paper examines the promise and policy of driverless cars*. Brookings. https://www.brookings.edu/articles/new-paper-examines-the-promise-and-policy-of-driverless-cars/

Karsten, J., & West, D. (2017, January 30). *Semi-autonomous vehicles must watch the road and the driver*. Brookings. https://www.brookings.edu/articles/semi-autonomous-vehicles-must-watch-the-road-and-the-driver/

Lin, P. (2015, December 8). *The ethical dilemma of self-driving cars - Patrick Lin*. TED-Ed. https://ed.ted.com/lessons/the-ethical-dilemma-of-self-driving-cars-patrick-lin

NHTSA. (2021). *Automated vehicles for safety. NHTSA*. Www.nhtsa.gov; National Highway Traffic Safety Administration. https://www.nhtsa.gov/vehicle-safety/automated-vehicles-safety

Pariser, E. (2025). *Beware online "filter bubbles."* Ted.com; TED Talks. https://www.ted.com/talks/eli_pariser_beware_online_filter_bubbles/transcript(ExternalLink)/transcript

Villasenor, J. (2014, April 24). *Products liability and driverless cars: Issues and guiding principles for legislation*. Brookings. https://www.brookings.edu/articles/products-liability-and-driverless-cars-issues-and-guiding-principles-for-legislation/